Pre-Assignment: Advanced RAG Masterclass

Objective

Assess participants' readiness for an advanced, production-focused RAG (Retrieval-Augmented Generation) program and establish a common baseline.

Part A: Conceptual Questions

1. Explain the complete lifecycle of a RAG system.

Cover:

- Data Ingestion
- Chunking
- Embedding Generation
- Vector Storage
- Retrieval
- Prompt Construction
- LLM Response Generation

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2. What are the limitations of a naïve RAG pipeline?

Discuss at least five challenges such as:

- Poor retrieval quality
- Hallucinations
- Context window limitations
- Chunking issues
- Latency and scalability concerns

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3. Compare the following retrieval approaches:

- Dense Retrieval
- Sparse Retrieval (BM25)
- Hybrid Retrieval

For each, explain:

- How it works
- Advantages
- Limitations
- Suitable use cases

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4. Explain how embeddings work.

Include:

- Vector representations
- Semantic similarity
- Cosine similarity
- Impact of embedding model selection on retrieval quality

5. What factors influence retrieval performance in a RAG system?

Discuss:

- Chunk size
- Chunk overlap
- Embedding model
- Top-K retrieval
- Metadata filtering
- Reranking

Part B: Hands-on Assignment

Build a Basic RAG Pipeline

Using any document collection (PDFs, reports, articles, research papers, etc.), create a working RAG system.

Minimum Requirements

1. Load and process documents.
2. Create chunks using an appropriate chunking strategy.
3. Generate embeddings.
4. Store embeddings in a vector database.
5. Retrieve relevant chunks for a query.
6. Generate answers using an LLM.